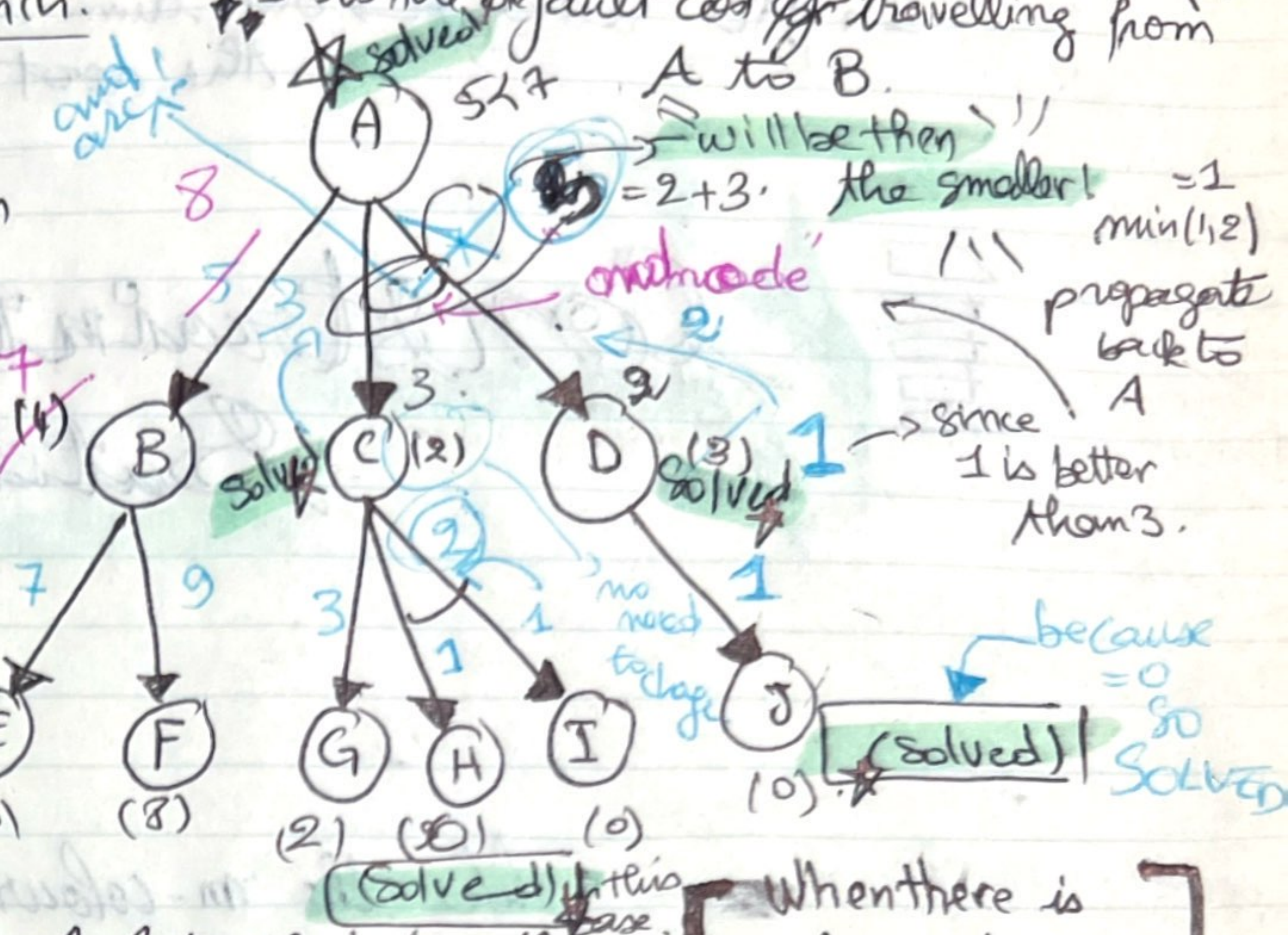
AND-OR* Search.

AO* Search:

divide the problem to smaller sub-problems handled in parts.

1 is the default cost for travelling from A to B.

Since we got the root is solved → we solved the problem



Step 01: → We calculate the best path from the node A.

$\to f(A-B) = g(B) + h(B) = 1+4 = \boxed{5}$

When there is the and arc we calculate the sum of all of them.

We have and arc! $\to f(A-C, D) = g(C) + h(C) + g(D) + h(D) = 3 + 2 + 1 + 1 = \boxed{7}$

Step 02: choose the best among E and F:

$f(B-E) = 6+1 = 7.$ → best because it's less

$f(B-F) = 8+1 = 9$

→ We go up again and update the heuristic value of B:

$* \to \boxed{7}$

Since the sub-problem is solved and got its real value.

f(A) must be calculated again